

Medical abbreviation bake-off

CS496 Project 0 | DRAFT - real benchmark not run

1. Task and data

Expand one abbreviation in a short UK medical-record context. Example: "The nurse recorded BP before the consultation."
Expected: blood pressure. The prompt supplies a fixed 60-term output vocabulary but never supplies the abbreviation-to-answer mapping.

50 frozen test items and 10 development items use disjoint abbreviations. Factual expansions come from the NHS glossary; contexts are original synthetic sentences and contain no patient data. Two-person human label review status: pending.

2. Experiment

top_api: gemini-3.1-pro-preview

cheap_api: gemini-3.5-flash-lite

local: qwen2.5:1.5b

Same literal prompt, vocabulary, question order, parser, temperature 0 and output limit 64 for all models. One call per item; sequential requests; no retries, retrieval, tools or chat history. Blank, non-vocabulary, extra-text or multiline outputs fail parsing; outer whitespace alone is ignored. Refusals, timeouts, truncation and API failures count as wrong.

Latency is wall time from send to full response for every attempt; p50/p95 use linear interpolation. The first local cold load is included. Local generation tokens/second excludes prompt processing. Exact model/version, UTC date, file hashes and local model digest are saved in run metadata.

3. Results

Model	Correct	p50 / p95 ms	USD / 1k	P / R / T / O
gemini-3.1-pro-preview	pending	pending	pending	pending
gemini-3.5-flash-lite	pending	pending	pending	pending
qwen2.5:1.5b	pending	pending	pending	pending

P=parse errors; R=refusals; T=timeouts; O=other execution errors. All remain in the accuracy denominator. Missing usage means unknown cost, never zero. Final numbers come directly from results/per_item.csv.

Evidence, cost and decision

4. Wrong answers (first three per model)

gemini-3.1-pro-preview - pending real run

gemini-3.5-flash-lite - pending real run

qwen2.5:1.5b - pending real run

5. Cost, traffic and hardware

API cost uses real usage token counts and paid standard USD list prices checked 2026-09-11. Local cost uses measured request throughput plus the hardware/electricity and operator-time assumptions in config.json. Baseline traffic is 1,000 requests/month and 100x traffic is 100,000/month.

Numeric 100x and break-even values are pending the real run and user-supplied local cost assumptions. Break-even is only defined when the API variable cost exceeds the local variable cost.

Hardware details and local cost are pending the final run on the student machine.

6. Choice and limitations

Decision pending real results. Predeclared rule: choose the lowest-cost model reaching at least 95% exact-match accuracy and p95 below 10 seconds; if none qualifies, report that none meets the target.

This vocabulary-constrained benchmark measures canonical matching, not clinical competence. The contexts and answer vocabulary may make the task easy. With 50 test cases, one item changes accuracy by two percentage points. Final test items must never be removed or repaired after seeing model outputs.

Sources: NHS abbreviation glossary; Google Gemini model/pricing documentation; Ollama Qwen2.5 model metadata. Full methodology and URLs are in data/labelling_note.md and README.md.